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InvestSmart

An AI-Powered Entrepreneurial Simulation & Decision
Support Platform

Software Project Management Proposal

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1. Project Title

InvestSmart—An AI-Powered Entrepreneurial Simulation & Decision Support Platform

2. Introduction & Problem Statement

In Bangladesh, a large segment of youth and students faces unemployment or underemployment, despite possessing strong aspirations to become self-reliant. Many express interest in starting their own businesses, yet the absence of practical business knowledge, market insights, and hands-on experience prevents them from taking informed steps. Identifying viable business opportunities, understanding market demand, and evaluating risks remain significant challenges for beginners.

At the same time, business students often struggle to practically apply the theoretical knowledge gained from their academic curriculum. There is a clear need for an accessible, simulation-based learning platform that can bridge the gap between classroom concepts and real-world entrepreneurial decision-making.

InvestSmart is an AI-powered web platform designed to address these challenges through a unified solution. The platform serves **dual purposes**:

- **Educational support for business students**, enabling them to apply theoretical concepts in realistic, interactive business simulations.
- **Decision-making assistance for aspiring entrepreneurs and unemployed youth**, helping them evaluate business ideas, understand investment requirements, and explore market feasibility with confidence.

These problems show why we need a platform that is easy to reach and interactive. It should use AI to help out. That way, it can give practical skills to those starting businesses and to business students. It would guide their decisions and offer simulations from real life.

3. Motivation & Rationale

Bangladesh's youth face a growing challenge: rising unemployment, limited access to practical business knowledge, and a lack of guidance to start sustainable ventures. Although many are eager to become self-reliant, they struggle to identify viable business opportunities, as-

sess market demand, or understand investment risks. At the same time, business students remain unable to translate theoretical lessons into real-world decision-making, creating a critical gap between education and entrepreneurship.

InvestSmart is designed to address these core problems. By combining AI-driven insights with interactive business simulations, the platform makes complex entrepreneurial decisions understandable and accessible. It empowers students and aspiring entrepreneurs to analyze markets, test business ideas, forecast profits, and evaluate risks—before investing any resources.

The initiative directly advances key Sustainable Development Goals:

- **SDG 4 – Quality Education:** offering hands-on, applied learning tools.
- **SDG 8 – Decent Work & Economic Growth:** enabling youth to pursue informed self-employment and entrepreneurship.
- **SDG 9 – Industry, Innovation & Infrastructure:** building a digital innovation that strengthens the entrepreneurial ecosystem.
- **SDG 17: Partnerships for the Goals:** Establishes strategic partnerships with business faculties and departments across universities

InvestSmart’s motivation is simple: **equip Bangladesh’s youth with the practical skills, insights, and confidence needed to make informed economic decisions and create sustainable livelihoods.**

4. Background Study / Related Works

Global research consistently shows that **simulation-based learning and AI-driven guidance** are transforming entrepreneurship education. Business Simulation Games (BSGs) have proven highly effective in helping learners practice real-world decision-making—such as pricing, investment planning, market analysis, and risk evaluation—within a safe, risk-free environment. Studies highlight that students who use simulations show **higher engagement, stronger confidence, and better entrepreneurial intentions** compared to those taught through traditional theory-only methods.

Recent developments in generative AI have further elevated this field. AI-powered platforms now provide **personalized coaching, dynamic market scenarios, and adaptive feed-**

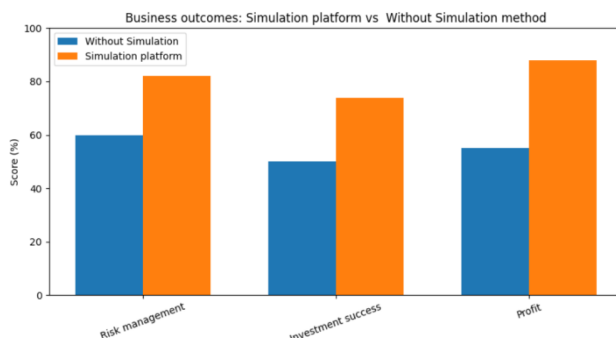


Figure 1: Experiment result on simulation business study

back, enabling learners to convert ideas into actionable business strategies with remarkable clarity. These advancements demonstrate a global shift toward technology-enabled experiential learning—yet such solutions remain largely absent in Bangladesh.

This gap presents a powerful opportunity: **InvestSmart** brings internationally validated methods—simulation, AI mentoring, and experiential learning—into a unified platform tailored for Bangladesh’s youth and students.

4.1 Key Resources

- Tan R.F. et al. — *Experiential Learning with Business Simulations* (2023)
- MDPI (2023) — *Simulation Games in Sustainable Learning*
- Zhu & Luo (2025) — *AI-Empowered Business Plan Systems*
- Mollick et al. (2024) — *AI Agents for Scalable Educational Simulations*

4.2 Onspot study

This project is also informed by insights collected directly from real Bangladeshi entrepreneurs and small business investors. Through interviews and observation, we noted their challenges in planning, pricing, marketing, and managing early-stage risks. Each step of their business journey successes, failures, and decision patterns was documented as ground-truth data. These practical experiences form the foundation for designing a realistic and locally relevant business simulation platform.

5. Proposed Solution & Technical Methodology

5.1 Proposed Solution

InvestSmart is an AI-powered web platform designed to help Bangladeshi youth, students, and aspiring entrepreneurs learn, test, and validate business ideas through interactive simulations. The platform combines **real-time market insights, AI-driven guidance, and hands-on business practice** to bridge the gap between academic theory and real-world entrepreneurship.

The solution provides:

- **Business Idea Simulator:** Users test business concepts in a virtual environment with realistic market conditions.
- **AI Business Mentor:** A generative AI assistant that provides step-by-step guidance on planning, budgeting, pricing, marketing, and risk assessment.
- **Market Feasibility Analyzer:** Automated evaluation of demand, competition, investment needs, and profit potential.
- **Learning Hub:** Short modules, case studies, and tutorials to strengthen practical business skills.
- **Progress & Performance Tracking** – Analytics dashboards showing improvement across decisions and simulations.

InvestSmart empowers beginners to make informed decisions *before* investing money, directly supporting youth empowerment, self-employment, and SDG-aligned development.

5.2 Technical Methodology

5.2.1 AI Model / Approach

a. Retrieval-Augmented Generation (RAG):

- Used for fetching relevant business cases, examples, and learning content.
- Combines vector similarity search with LLM generation to produce personalized recommendations and explanations.

- **Example Use-Cases:**

- AI Business Mentor retrieves topic-specific content from structured knowledge base.
- Business Idea Simulator fetches similar historical business cases to predict outcomes.

b. Large Language Models (LLM):

- Models like GPT-4, Mistral, or LLaMA used for:
 - Generating step-by-step guidance and actionable advice.
 - Explaining complex business concepts in simple terms.
 - Suggesting mini-case exercises or implementation examples.
- Fine-tuned or prompt-engineered for domain-specific business knowledge.

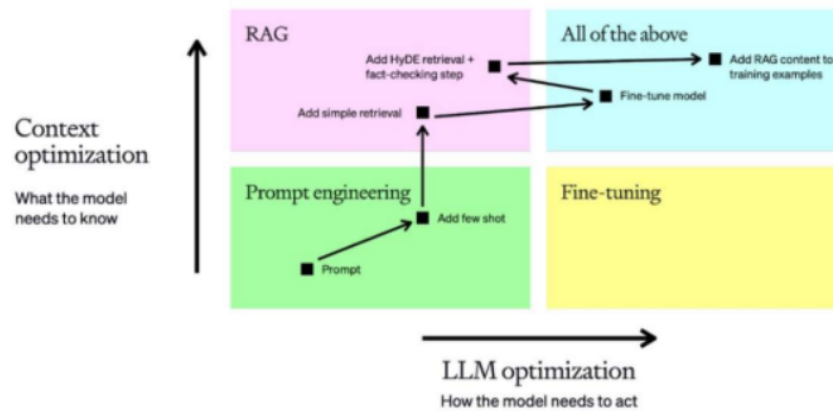


Figure 2: AI based solution approach

5.2.2 System Design & Technical Workflow

Overall Flow:

1. User Input Layer:

Collects business idea, budget, location, team experience, or learning preferences.

2. Data Preprocessing & Storage:

Structured data: PostgreSQL (business cases, KPIs).

Unstructured data: MongoDB or vector database (learning modules, case studies).

3. AI Processing Layer:

RAG pipeline: Retrieve relevant content or past business cases.

LLM: Generate personalized insights, learning guidance, or scenario predictions.

Predictive ML models: Compute KPIs, risk scores, and feasibility metrics.

4. Simulation / Scenario Engine:

Evaluate multiple "what-if" scenarios for business ideas.

Generate outputs like projected revenue, risk assessment, and success probability.

5. Learning & Guidance Layer:

AI Mentor provides step-by-step explanations, practical examples, and gamified exercises.

6. Analytics & Dashboard:

Track user progress, learning outcomes, and simulation performance.

Visualize improvement and recommendations via charts and dashboards.

5.2.3 Datasets**a. Internal / Proprietary Datasets:**

- Structured business case experiments, KPI results, success/failure scenarios.
- User interaction and learning progress data (for analytics and model feedback).

b. External Datasets:

- Market trends, competitor data, and economic indicators via web scraping (BeautifulSoup, Scrapy) or APIs (Google Trends, World Bank, Kaggle datasets).
- Business tutorials, books, and online resources for RAG-based learning content.

c. Data Processing:

- Text data → cleaned, tokenized, embedded into vector space for RAG.
- Numeric/market data → normalized and structured for predictive modeling.

5.2.4 Tools & Frameworks

Layer	Tools / Frameworks
AI / RAG	LangChain

LLM	Free models
ML/Analytics	Scikit-learn, Pandas, NumPy, Plotly, Matplotlib
Backend	Python, Django
Frontend	React.js
Data Collection	BeautifulSoup, Scrapy, APIs
Database	PostgreSQL, vector databases for embeddings
Deployment	Docker, Github, Vercel

6. Scope and Pilot Implementation

6.1 Scope

InvestSmart focuses on helping Bangladeshi youth learn, test, and validate business ideas using AI-driven tools. The project scope includes:

- **Business Idea Simulator:** Virtual testing of business ideas with cost, demand, risk, and competition factors.
- **AI Business Mentor:** Step-by-step guidance on planning, budgeting, pricing, and marketing using RAG + LLMs.
- **Market Feasibility Analyzer:** Automated scoring of demand, competition, investment needs, and profitability.
- **Learning Hub:** Short modules and case studies that build practical business skills.
- **Progress Dashboard:** Tracks user improvement and decision quality.
- **Target Users:** Students, young entrepreneurs, and unemployed youth seeking self-employment.
- **Technology Coverage:** RAG for retrieval, LLMs for guidance, ML models for risk and feasibility prediction.

6.2 Pilot Implementation

The pilot will include a small set of core features to test the basic usefulness of InvestSmart.

Pilot Features

Business Idea Input Users add a business idea with basic cost and price info.

Basic Simulation Shows estimated profit and risk level.

AI Mentor (Simple Tips) Gives short suggestions on pricing, cost, and marketing.

Feasibility Score Provides a basic rating: High / Medium / Low.

Small Learning Section Only 2–3 quick lessons on pricing, customers, and risk.

Simple Progress View Shows number of simulations and completed lessons.

7. Budget & Resource Plan (6 Months)

7.1 Recurrent Expenditure (6 Months)

Code	Description	Unit	Cost per Unit (BDT)	Estimated Cost (BDT)	Justification
A1	Full Stack Developers (4)	4	70,000/month × 4	280000*6	Core system + module development
A2	ML Engineer (part-time)	1	50,000/month × 6	3,00,000	AI model & feasibility
A3	UI/UX Designer (part-time)	1	25,000/month × 4	1,00,000	Frontend + dashboard
A4	Project Coordinator	1	18,000/month × 6	1,08,000	Documentation, reporting
A5	SME Domain Advisor	1	12,000/session × 4	48,000	Business insights
A6	Cloud Hosting (VPS)	1	5,000/month × 6	30,000	Hosting + storage
A7	Workshops/Training	2	25,000/event	50,000	Youth/student training

A8	Publication + Conference	1	35,000	35,000	Journal + showcase
A9	Intern/Student Assistant	1	10,000/month $\times 4$	40,000	Testing + annotation
A10	Field Survey/Data Validation	-	Lump 35,000	35,000	Dataset validation
A11	Team Refreshments/Meetings	-	5,000/month $\times 6$	30,000	Team coordination
A12	LLM API Usage	-	Lump Sum	1,00,000	AI inference + API cost
Total Recurrent Cost =					24,56,000 BDT

7.2 Capital Expenditure (One-Time)

Code	Description	Unit	Cost per Unit (BDT)	Estimated Cost (BDT)	Justification
B1	Developer Laptop	5	70,000	70,000 $\times 5$	Development + ML
B2	Testing Smartphone/Tablet	1	25,000	25,000	UI testing
B3	Cloud Storage & Backup	1	35,000	35,000	Backup repository
B4	GPU/Compute (Colab Pro+/Kaggle)	1	35,000	35,000	Model training
B5	Learning Material Toolkit	1	25,000	25,000	Training content
B6	Office Accessories (Printer/UPS)	1	30,000	30,000	Documentation support
Total Capital Cost =					2,30,000 BDT

7.3 Operational / Contingency

Code	Description	Estimated Cost (BDT)	Justification
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C1	Contingency (approx. 25% of total)	2,50,000	Unforeseen expenses
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7.4 Grand Total Budget (6 Months)

Category	Amount (BDT)
Total Recurrent	2736000
Total Capital	2,30,000
Total Contingency	2,50,000
Grand Total (Final)	32,16,000 BDT

8. Impact on Society

InvestSmart will create high-value social and economic impact in Bangladesh through:

8.1 Youth Empowerment & Employment

- Allows young people to try out business ideas with minimal risk.
- Reduces fear and financial risk while exploring entrepreneurship.

8.2 Improved Business Education

- Students gain an applied understanding of finance, marketing, and operations.
- Enhances employability and entrepreneurial skill development.

8.3 Improved survival rates of startups

- AI-driven feasibility evaluation reduces unplanned losses.
- Data-driven decisions increase the sustainability of SMEs.

8.4 Inclusive Access

- A low-cost and scalable solution that universities, vocational institutes, unemployed youth, and community entrepreneurs can use.

9. Alignment with Sustainable Development Goals (SDGs)

InvestSmart directly contributes to:

9.1 SDG 4: Quality Education

- Transforms abstract academic learning into tangible simulation-based experiences.
- Promotes active learning with measurable performance outcomes

9.2 SDG 8: Decent Work & Economic Growth

- Facilitates entrepreneurship and small-business formation
- Encourages youth-led economic activities, thereby increasing productivity.

9.3 SDG 9: Industry, Innovation & Infrastructure

- Leverages AI to support industrial innovation, SME growth, and digital transformation

9.4 SDG 17: Partnerships for the Goals

- Establishes strategic partnerships with business faculties and departments across universities
- Enables collaboration between academia and technology to bridge theoretical-practical gaps
- Creates institutional partnerships for student access to applied learning tools
- Facilitates knowledge sharing between business education institutions and AI-driven platforms
- Supports multi-stakeholder engagement for sustainable entrepreneurship development

10. Revenue & Sustainability Plan

To ensure long-term financial viability and continuous platform improvement, the simulation system will operate with a diversified monetization strategy.

10.1 Individual Users – Freemium + Subscription Hybrid Model

The business model for independent learners will follow a structure similar to modern SaaS tools:

10.1.1 Freemium Tier (Free Access)

Users can access:

- Limited simulations (e.g., 5 simulation runs per month)

10.1.2 Premium Subscription Tier

After exceeding free usage, users will be encouraged to subscribe.

Subscription options:

- Monthly subscription
- 6 months subscription
- Annual subscription (discounted)

Premium users receive:

- Unlimited simulation access
- Expanded business scenarios and market variations
- Personalized AI-driven improvement feedback
- Priority support

10.2 Institutional Licensing Model

For academic institutions and corporate training teams, a licensing-based subscription will be used.

10.2.1 Pricing Model:

Institutions will subscribe using any of the following:

- Semester-Based License
- Annual License
- Enterprise packages for large cohorts

Pricing will be based on:

- Number of students
- Feature selection (basic / advanced)
- Number of course facilitators

11. Expected Outcomes & Deliverables

11.1 Deliverables for Pilot (Course Submission):

- AI Feasibility Study Module (prototype working)
- Business Basic Simulation Scenario
- Market Insight Dashboard: for 2-3 sectors
- Web-based working interface (demo)
- Documentation + short demo presentation

11.2 Long-term Outcomes:

- Fully featured localized simulation lab for entrepreneurship
- Adoption by academic institutions
- Increased youthful attempts in startups with reduced risk
- Enhanced financial literacy in Bangladesh

12. Risks & Mitigation Strategies

Risk	Description	Mitigation
Limited Local Data	Lack of structured datasets	Use proxies, public data, and curated SME case studies
Model Accuracy Concerns	Uncertainty in forecasts	Conservative outputs, confidence ranges, and explainability
User Adoption Challenges	Users may distrust AI	Clear guidance, improved UI/UX, educational onboarding
Scalability Issues	Increases with user load	Modular architecture, containerization, cloud scaling

13. Assessment Mapping

This proposal meets the evaluation expectations by:

- Clearly identifying a real, practical problem in Bangladesh
- Proposing a novel AI-powered solution with a realistic pilot scope
- Providing a technically feasible architecture
- Showing strong alignment with SDGs
- Demonstrating measurable academic and economic impact

14. Conclusion

InvestSmart is more than a theoretical solution; it bridges the gap between ambition and actionable entrepreneurship. By combining simulation, AI-driven analysis, and practical learning, it empowers youth and students to enter the business world with confidence, clarity, and capability. The pilot implementation will demonstrate core functionalities and lay the foundation for a full platform that can drive innovation, economic growth, and sustainable development within Bangladesh.